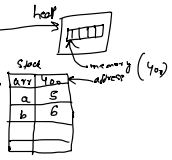
Arrays 2

02 April 2026 08:05

```
main(){
  int[] arr = new int[5];
  int a=5;
  byte b=6;
}
```



[0, 1, 2, 3, 4]
[1, 2, 3, 4, 5]

arr[0] → 1

Sout(arr[3]) → 4

arr.length → 5

[1, 2, 3, 4 - - - - - - -]
0 1 2 3 - - - - - - - arr.length-1

→ Traverse

```
for(int i=0; i< arr.length ; i++){
    Sout(arr[i]);
}
```

→ 1 2 3 4 5

int[] arr = new int[5];

Print

```
for(int i=0; i< arr.length ; i++){
    Sout(arr[i]);
}
```

→ 0 0 0 0 0
→ 1

Java by default arr → 0

arr[0] = 1;

user input of arrays.

```
Scanner Sc = new Scanner();
int n = Sc.nextInt();
int[] arr = new int[n];
for(int i=0; i<n ; i++){
    arr[i] = Sc.nextInt();
}
```

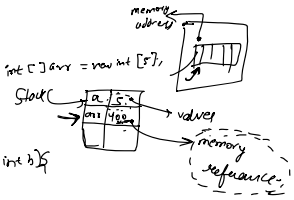
0, 1, 2, ... n-1

//Print.

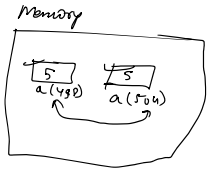
Imp!!

int a = 5; → declaration

int[] arr = new int[5];



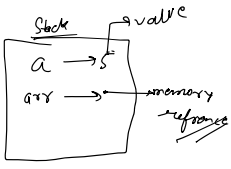

```
int all (int a, int b){
    a = a+1;
    return a+b;
}

void main(){
    int a=5;
    int b=6;
    int sum = add(a,b)
    Sout(a);
}
```

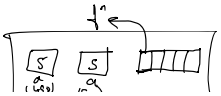


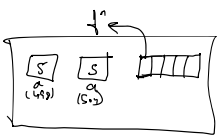

```
void change (int[] arr){
    arr[0] = 5;
}

main(){
    int[] arr = {1,2,3,4,5};
    change(arr);
    //Print
    for(int i=0; i< arr.length; i++){
        Sout(arr[i]);
    }
```

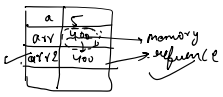
[1, 2, 3, 4, 5]
→ 5

→ [5, 2, 3, 4, 5]









Q. Find the Second largest number in an array (Don't use Sorting).

largest = 9, Sec_largest = 8

[1, 2, 9, 6, 8]

```
int largest = arr[0], Sec_largest = arr[0];
for(int i=0; i< arr.length; i++){
    if(arr[i] > largest){
        → Sec_largest = largest;
        → largest = arr[i];
    }
    else if( arr[i] > Sec_largest && arr[i] < largest){
        Sec_largest = arr[i];
    }
}
```

Director (1st)
Boss (2nd)
Manager (3rd)
Employee

[1, 2, 3, 4, 5] (asc) — or — [5, 4, 3, 2, 1] (dec)

Q. Sorted array, you have find a repeating number (if have). → only one number repeats.

[1, 1, 2, 3, 4]

[1, 2, 3, 3, 5, 6]

xor = 1 ^ 2 ^ 3 ^ 3 → 0

xor = 1 ^ 2

[1, 2, 3, 3, 4, 5]

```
int duplicate = -1;
for(int i=0; i< arr.length-1; i++){
    if(arr[i] == arr[i+1]){
        duplicate = arr[i];
    }
}
→ Sort (duplicate);
```

i → 0 → arr.length

0 1 2 3 4 5

i=0 arr[0] = arr[1]

i=1 arr[1] = arr[2]

⋮

i=4 arr[4] == arr[5]

0, 1, 2, 3

Q. Find the number which is not repeating

[1, 2, 1, 3, 4, 3, 4]

i → j, i → j → o/p → 2

i=0 arr[0] == arr[1]

arr[1] == arr[2]

arr[3]

[1, 2, 1, 3, 4, 3, 2, 1]

i ← j → o/p → 2

i = 0

i = 1

i = 2

i = 3

arr[0] == arr[6]

arr[1] == arr[2]

arr[2] == arr[3]

arr[3] == arr[4]